Grace — Black-hole entropy as record-count (#72): reconnect + T2543 sharpening (2026-08-06, K1238)

Fresh topic off the n=5 treadmill (Casey rebalance). My lane: boundary geometry. FIRST MOVE = corpus-reconnect (standing order), and it paid off immediately.

Reconnect: BST already has this (and already rejected my first instinct)

`notes/BST_Bekenstein_Quarter_Disambiguation.md` (Casey + Opus 4.6, March 13 2026) settled the Bekenstein 1/4 three ways and picked a winner: - Candidate A WINS: 1/4 = (1/2)_hol × (1/2)_Z2. Both factors are SO(2) structure. - **(1/2)_hol** — Bergman modes are holomorphic (p^+ = half of p_C). Rigorous (Bergman space theory). - **(1/2)_Z2** — charge conjugation C: m→−m = center −1 ∈ SO(2) on the Shilov S^1. Open (March Sec 13, problem 1: derive it as a *gauge* redundancy at saturation). - Candidate C (1/rank^2 = 1/4) — REJECTED as numerological (no rep-theory derivation; would give wrong answer at other ranks).

★ Self-catch (both directions, the discipline working)

My pre-read instinct for "does the boundary hand you the 1/4" was 1/4 = 1/rank^2. That is *exactly* the March note's rejected Candidate C. Corpus-reconnect caught it before I re-proposed a rejected numerological match — the same failure mode as this morning's off-by-one, caught this time *before* it hit the board. Owned.

What's genuinely new since March (three moves, tier-honest)

1. Robustness to today's crux. Both factors of the winning 1/4 live in the SO(2) — the "2" in SO(n,2) — which is identical in SO(4,2) and SO(5,2). The Bekenstein 1/4 lives entirely in the "2", never the "n". So the result is robust to the off-by-one crux (why SO(5,2) not SO(4,2)? — all in the *n*). The BH entropy does not wait on the n_C=5 forcing question.
2. Row-2 tie-in. The SO(2) I exhibited this morning as complex-structure/time-circle is the *same* SO(2) whose ±1 weight-split gives (1/2)_hol and whose center Z2 gives (1/2)_Z2. One object, three roles (time, holomorphic-half, charge-conjugation-half).
3. T2543 sharpens the open factor. March's undefined "committed contact configuration at Haldane saturation" = a T2543 record = holomorphic idempotent. A record is holomorphic (→ (1/2)_hol) and carries a matter/antimatter C-conjugate (→ (1/2)_Z2). The (1/2)_Z2 is then the no-hair theorem in record language: matter and antimatter records are C-conjugate; no-hair makes them horizon-indistinguishable; the C-orbit {e, Ce} collapses to one distinguishable state → factor 1/2. So (1/2)_Z2 is grounded in imported no-hair physics (GR-style external input, like G in GR) — not a pure-internal derivation, but no longer fully open.

Tier (honest)

Reconnection + framing/LEAD. NO new bank, no new theorem. Candidate A stands where March left it, now with: - (1/2)_hol — rigorous (unchanged). - (1/2)_Z2 — grounded in no-hair + T2543 (imported-derived), upgraded from "open" to "rides an imported theorem." Still not pure-internal; do not claim S_BH=A/4 as "Derived from D_IV^5 alone." - The whole 1/4 — robust to the n_C=5 off-by-one (new, useful given today).

Handoff (Elie toy = March open problem #2)

Evaluate the Bergman kernel restricted to the horizon Σ and verify the mode density dN/dA = 1/(4 ℓ_P^2) directly — a direct spectral-density check of Candidate A, independent of the factor-decomposition argument. Testable, concrete, uses the boundary geometry.

Extension (wide, safe): the 1/4 is UNIVERSAL across horizon types — and that's a consistency check on "it's the SO(2)"

Black-hole (Bekenstein-Hawking), cosmological (Gibbons-Hawking de Sitter), and Rindler/entanglement horizons all carry the same 1/4 — textbook. In the record picture this is not a coincidence: every horizon is a record-saturation boundary (the edge of what can be committed in a causal patch), and the two halvings are (1/2)_hol + (1/2)_Z2, both SO(2) structure — which is identical for *any* horizon because it's the domain's isotropy factor, not anything black-hole-specific. So the observed universality of 1/4 across horizon types is a consistency check that the 1/4 is the SO(2), not a solution-specific accident (this is exactly what killed Candidate B, which was black-hole-geometry-specific and couldn't survive Kerr/dS).

Scope discipline (explicit): this claims the coefficient (1/4) is universal and SO(2)-sourced. It says nothing about the area — for the dS horizon the area carries Λ, which is [[lambda_overdetermination_retracted]] structural-not-derived (K741). The 1/4 and the area are separate objects; I claim only the former. Do not let "dS entropy" drift into a Λ derivation.

Note

Pairs with T2543 (record=idempotent), Row 2 (SO(2)→time), and the universal-horizon 1/4 above. Nothing pushed.